Movement Disorders
Vol. 10, No. 5, 1995, pp. 678–684

Communications

Are Death Certificates Reliable to Estimate the Incidence of Parkinson's Disease?

To the Editor:

Studies of possible clusters and review of causation of Parkinson's disease (PD) have been correlated previously with exposure to herbicide or other potentially noxious agents in particular regions (1–3), with both internal and external toxins suggested as causal in PD. We investigated whether death certificates could be a reliable source for determining the presence of PD.

Methods and Results

A request was placed to the Central Ohio Parkinson Society for names of any patient who died from 1989 until 1992, who had been diagnosed and treated for PD, and had carried this diagnosis for several years preceding death. Two hundred eleven names were identified; only 155 (74%) of their original death certificates could be obtained. These death certificates were scanned for any mention of PD as the cause of the death under the columns "cause of death" or "other significant conditions." Ninety-nine of the 155 (63.9%) had PD listed in one section or the other (Table 1).

Discussion

It is tempting to use death certificates to estimate the incidence and prevalence of disease because certificates are in writing, represent a permanent record, and offer an unequivocal endpoint. Nevertheless, there are obvious problems with research that uses death certificates including potential lack of adequate physician input or interest; notation of dramatic terminal events but no mention of contributory causes; and, as we discovered, surprising difficulty in retrieving the files. Despite difficulties, death certificates have been used to estimate the incidence of disease in many diseases, including fetal death, cardiovascular disorders, and degenerative diseases (4–10). In our opinion, it is doubtful, short of a major local epidemic, that a slight increase in the incidence of PD will be detected by the use of death certificates.

TABLE 1. *Patients with Parkinson's disease (PD) who died 1989–1992, and percent of death certificates (DCs) listing PD*

Year	Total deaths	DCs Found	% Found	PD listed	% with PD listed
1989	47	29	61.7	23	79.3
1990	48	36	75.0	20	55.6
1991	46	34	73.9	22	64.7
1992	70	56	80.0	34	60.7
Total	211	155	73.5	99	63.9

The families of the 211 patients were all confident that the patients had PD, because the patient was being treated for it and had been so diagnosed up until the time of death. Despite this, 56 (36%) did not have PD listed on the death certificate as either a primary or secondary cause, or even as a disease condition present at death.

Conclusion

In central Ohio, death certificates often do not reflect the opinion of families that PD was present in the deceased relative during life. There are many possible explanations for this apparent deficiency, but there is only one probable conclusion: We cannot trust death certificates to reflect the accurate incidence of PD, at least not in central Ohio.

George W. Paulson
William M. Gill
Department of Neurology
Ohio State University
Columbus, Ohio, U.S.A.

References

1. Koller W, Vetere-Overfield B, Gray C, et al. Environmental risk factors in Parkinson's disease. *Neurology* 1990;40:1218–1221.
2. Rybicki BA, Johnson CC, Uman J, Gorell JM. Parkinson's disease mortality and the industrial use of heavy metals in Michigan. *Mov Disord* 1993;8:87–92.
3. Rajput AH, Utti RJ, Stern W, et al. Geography, drinking water, chemistry, pesticides and the etiology of Parkinson's disease. *Can J Neurol Sci* 1987;14:414–418.
4. Stegmayr B, Ospland K. Measuring stroke in the population: quality of routine statistics in comparison with a population based stroke registry. *Neuroepidemology* 1992;11:204–213.
5. Iso H, Jacobs DR, Goldman L. Accuracy of death certificate diagnosis of intracranial hemorrhage and non-hemorrhage stroke. *Am J Epidemiol* 1990;132:993–998.
6. Johnson I, Britton J, Kinneau W, et al. Rising mortality from cryptogenic necrosing alveolitis. *Br Med J* 1990;301:1017–1021.
7. Mossa MA, Shafie MZ, Khogali MM, et al. Reliability of death certificate diagnosis. *J Clin Epidemiol* 1990;43:1285–1295.
8. Chio A, Magnani C, Oddenino E, Tolardo G, Schiffer D. Accuracy of death certificate diagnosis of amyotrophic lateral sclerosis. *J Epidemiol Community Health* 1992;46:517–518.
9. Teperi J. Multi method approach to the assessment of data quality in the Finnish Medical British Registry. *J Epidemiol Community Health* 1993;47:242–247.
10. Kirby RS. The coding of underlying cause of death from fetal death certificates: issues and policy considerations. *Am J Publ Health* 1993;83:1088–1091.